NPTEL Online Certification Course
Industrial Engineering and Operations Research
Assignment Number 1: Detailed Solution
Indian Institute of Technology Guwahati

Q.1. Who among the following is considered the first military engineer?

(A) John Smeaton (B) Archimedes (C) Isaac Newton (D) Hans Christian Ørsted

Sol. Archimedes (287–212 BC) is credited as the first military engineer. Hence, option (B) is correct.

Q.2. Who devised a Chimney Mill in Britain in 1782 and a watermill in 1767?

(A) John Smeaton (B) L.A. Pelton (C) James Francis (D) F.W. Taylor

Sol. John Smeaton contributed to Mechanical Engineering by devising a Chimney Mill in Britain in 1782 and a watermill in 1767. Hence, option (A) is correct.

Q.3. F.W. Taylor's active contribution falls in the period

(A) 1750–1850 (B) 1851–1930 (C) 1931–1960 (D) 1961–2010

Sol. F.W. Taylor's active contribution falls in the period 1851–1930. His contribution was on method study, distinct identity of industrial/production engineering, use of electrical power, steel production, statistical quality control and queuing theory. He obtained his bachelor in mechanical engineering degree in 1883. His active contribution started from that year and continued up to till his death in 1915. Hence, option (B) is correct.

Q.4. When did Operations Research (OR) formally develop?

(A) 1939–1945 (B) 1914–1918 (C) 1947 (D) 1925

Sol. Operations Research (OR) was formally developed **during World War II (1939–1945)** as military forces sought **scientific methods** to enhance decision-making and resource allocation. Hence, option (A) is correct.

Q.5. Which of the following is/are an objective/s of Production Planning and Control?

(A) Reducing costs (B) Improving product quality

(C) Efficient resource utilisation (D) Financial Accounting

Sol. The objectives of Production Planning and Control are Efficient resource utilization, Meeting delivery schedules, reducing costs, and improving product quality. Financial Accounting is not part of Production Planning and Control. Hence, options (A), (B) and (C) are correct.

Q.6. In a planning process, the duration of short-range planning is

Indian Institute of Technology Guwahati

(A) up to 3 months (B) 3 to 12 months (C) 12 to 24 months (D) more than 24 months

Sol. In a planning process, the duration of Long-range plans is over 1 year, Intermediate-range plans are 3 to 12 months, and Short-range plans are up to 3 months. Hence, option (A) is correct.

Q.7. Airlines adjust ticket fares based on factors like demand, weather, and festive seasons. This practice is called

(A) Aggregate Planning (B) Revenue Management

(C) Capacity Scheduling (D) Demand Forecasting

Sol. In revenue management, the idea is to match capacity and demand by charging different prices based on the customer's willingness to pay. The challenge is to identify those differences and price accordingly. Also referred to as yield management, it determines the allocation of resources to maximize revenue or yield. Hence, option (B) is correct.

Q.8. Capacity planning involves determining the required

(A) equipment (B) workforce (C) materials (D) sales

Sol. Capacity planning involves determining the resources (like facilities, equipment, workforce, and materials) needed to meet current and future demand, ensuring a business can produce or deliver what customers need. Sales determination is not the task of capacity planning. In fact, based on the expected sales, capacity planning is carried out.

Q.9. In the Pugh matrix for the concept selection technique, which of the following statement/s is/are true?

(A) Best concept will be one with the minimum 's' symbols.

(B) For concepts better than the datum concept, the '+' symbol is assigned.

(C) For concepts worse than the datum concept, the '−' symbol is assigned.

(D) 's' indicates a datum level concept.

Sol. The concept evaluation matrix compares generated design concepts against the iteration datum. For concepts better than the datum concept, the '+' symbol is assigned; for worse concepts, the '-' symbol is assigned. 's' indicates the datum level concept. The best concept will be the one with the maximum '+' symbols. Hence, options (B), (C) and (D) are correct.

Q.10. Which of the following is not a part of the proper sequence of design stages?

(A) Conceptual (B) Embodiment (C) Detailed (D) Manufacturing

Sol. Three stages of design are Conceptual design stage, Embodiment stage and Detailed design stage. Conceptual design stage takes the design problem statement and generates broad solution for it. Solution is in the form of design concepts. In Embodiment stage, concepts are converted into bodily form. Final check is made on function, spatial compatibility, design aesthetics and economics.

Indian Institute of Technology Guwahati

In Detailed design, the outcome of this stage is a set of drawings obtained by rigorous design calculations. Options (A), (B) and (C) are sequence of design stages. Hence, option (D) is correct.

Q.11. The primary purpose of a morphological matrix in a product design is to

(A) reduce product weight (B) generate and explore design alternatives

(C) finalize the best manufacturing method (D) To perform market analysis

Sol. In a morphological matrix, each axis represents a major function. Each method performing the functions is a co-ordinate along the axis. Each cell represents a potential design solution. Hence, option (B) is correct.

Q.12. In Cardinal utility, the unit of utility is

(A) dollar (B) number of units (C) utils (D) utilities

Sol. Cardinal utility theory assumes that utility (satisfaction from consuming goods) can be measured numerically, using units called "utils". Hence, option (C) is correct.

Q.13. For the Analytical Hierarchy Process (AHP), which of the following statement/s is/are true?

(A) It is a part of linear programming problems.

(B) It has an axiomatic foundation.

(C) It makes structured, multi-criteria decision-making easier.

(D) It performs regression analysis.

Sol. Analytical Hierarchy Process (AHP) is a structured technique used for organizing and analyzing complex decisions by breaking them down into a hierarchy of criteria and alternatives. It helps in setting priorities and making the best decision when both qualitative and quantitative aspects need to be considered. It has an axiomatic foundation which requires that the annihilation condition is not satisfied. Hence, option (B) and (C) are correct.

Q.14. In AHP, pairwise comparisons are typically rated using

(A) a 1–5 Likert scale

(B) a binary yes/no choice

(C) probabilistic weights

(D) a 1–9 scale of relative importance

Sol. In AHP, pairwise comparisons are typically rated on a scale of 1–9. Hence, option (D) is correct.

Q.15. Membership grades for the triathlon sports event in three sports are given in the table.

Name	Membership grade		
	Swimming	Bicycling	Running

Indian Institute of Technology Guwahati

Anurag	0.65	0.73	0.81
Rohan	0.77	0.85	0.91
Vikash	0.90	0.66	0.78
Harsh	0.84	0.93	0.69
Sourav	0.98	0.87	0.74

As per the compensating strategy, the ranking of athletes (from best to worst) is

(A) Anurag, Vikash, Harsh, Rohan, Sourav

(B) Rohan, Sourav, Vikash, Harsh, Anurag

(C) Harsh, Sourav, Rohan, Anurag, Vikash

(D) Sourav, Rohan, Harsh, Vikash, Anurag

Sol. An overall membership grade based on a compensating strategy is given by

$$\mu_{\text{overall}} = \left(\mu_1 \mu_2 \mu_3\right)^{\frac{1}{3}}$$

The last column of the table shows each athlete's overall membership grade. The athletes with the highest membership grade will have the highest rank. The ranking of athletes (from best to worst) is Sourav, Rohan, Harsh, Vikash, and Anurag. Hence, option (D) is correct.

Name	Membership grade			Overall Membership grade
	Physics	Mathematics	English	
Anurag	0.65	0.73	0.81	0.727
Rohan	0.77	0.85	0.91	0.841
Vikash	0.90	0.66	0.78	0.773
Harsh	0.84	0.93	0.69	0.813
Sourav	0.98	0.87	0.74	0.857

Q.16. The ratings of products A and B in a scale of 1 (poor) to 10 (Excellent) in two criteria are given in the table.

	A	B
Performance	8	7
Economy	6	9

If 38% importance is given to performance and 62% to economy, the scores of product A and B are respectively

(A) 7.6, 7.38 (B) 7.38, 7.86 (C) 7.38, 7.6 (D) 7.86, 7.4

Sol. Score of product A = (8 × 0.38) + (7 × 0.62) = 7.38

Score of product Y = (6 × 0.38) + (9 × 0.62) = 7.86

Hence, option (B) is correct.

Q.17. A carpenter typically takes 5 days to make one chair, working 7 hours per day. He charges ₹80 per hour. You hired him on a daily wage basis, paying him for 7 hours of work each day. However,

Indian Institute of Technology Guwahati

due to laziness and personal needs, he worked only 6 hours per day. What is the total amount you will end up paying the carpenter for making one chair?

(A) ₹ 3267 (B) ₹ 2800 (C) ₹ 3000 (D) ₹ 2333

Sol. Given,
Days to make a chair= 5
Working hours per day = 7
Hourly rate of the carpenter = ₹80 per hour
Total man-hours needed for making chair = 5×7=35

Carpenter works only for six hours in a day.

Hence, time to make the chair $= \frac{35}{6}$ days

He will charge per day based on seven hour's work. Hence, his per day earning is ₹80×7.

Hence, his total earning in making the chair (in rupees) = $\frac{35}{6} \times 80 \times 7 = 3266.67$

Hence, option A is correct.

Q.18. Setup cost for a machine is ₹40000. Variable cost for making one unit of product is ₹90. If the production cost per unit comes out to be ₹120, how many units are produced in a batch?

(A) 1333 (B) 1000 (C) 333 (D) 30000

Sol. Given,
Setup cost (fixed cost) = ₹40000
Variable cost per unit = ₹90
Production cost per unit = ₹120
Let the number of units produced in a batch be x.
Total cost= Setup cost + (Variable cost per unit × Batch Size)
$= 40000 + 90x$

$$\text{Total cost per unit} = \frac{40000 + 90x}{x}$$

$$\Rightarrow 120 = \frac{40000 + 90x}{x}$$

$$\Rightarrow x = 1333.34$$

Hence, option (B) is correct.

Q.19. Which type of production usually needs a process layout?
(A) Flow production (B) Batch production (C) Mass production (D) Job production

Sol. In a process layout, similar machines (e.g., all milling machines) are grouped together. This layout is typical in batch production, where products are made in batches and move through various departments based on the required operations. Therefore, correct option is (B).

Indian Institute of Technology Guwahati

Q.20. Which one of the following is not a characteristic of flow production?
(A) Low flexibility
(B) High flexibility
(C) High efficiency
(D) Significant initial investment

Sol. Flow production has the following characteristics: Low flexibility, high efficiency and significant initial investment. Hence, option (B) is correct.

Indian Institute of Technology Guwahati



NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati



Indian Institute of Technology Guwahati

NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

Indian Institute of Technology Guwahati



NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati



Indian Institute of Technology Guwahati

NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
<COURSE NAME>
<Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

Indian Institute of Technology Guwahati